CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

# Chapter 6: Triangles

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. State whether all rectangles are similar. Give a reason.

2. In $\triangle ABC$, $DE \parallel BC$ with $D$ on $AB$ and $E$ on $AC$. If $AD = 3$ cm, $DB = 4.5$ cm and $EC = 6$ cm, find $AE$.

3. In $\triangle PQR$, $E$ and $F$ lie on $PQ$ and $PR$ with $PE = 6$ cm, $QE = 7.5$ cm, $PF = 4$ cm and $RF = 5$ cm. Is $EF \parallel QR$?

4. In $\triangle ABC$, $DE \parallel BC$. If $AD = x + 1$, $DB = x$, $AE = x + 3$ and $EC = x + 1$, find $x$.

5. $\triangle ABC \sim \triangle PQR$ with $AB = 6$ cm, $BC = 8$ cm, $CA = 10$ cm and $PQ = 9$ cm. Find $QR$ and $RP$.

6. The sides of two triangles are 2.5 cm, 6 cm, 6.5 cm and 5 cm, 12 cm, 13 cm. Are they similar? Name the criterion.

7. The perimeters of two similar triangles are 30 cm and 20 cm. If one side of the first triangle is 12 cm, find the corresponding side of the second.

8. A vertical stick 12 m long casts a shadow 8 m long. At the same time a tower casts a shadow 40 m long. Find the height of the tower.

9. A girl 1.5 m tall stands 3 m from a lamp-post and casts a shadow 4.5 m long. Find the height of the lamp-post.

10. $ABCD$ is a trapezium with $AB \parallel DC$ whose diagonals meet at $O$. If $AO = 3$ cm, $OC = 5$ cm and $BO = 4.5$ cm, find $OD$.

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## Answer Key

1. No. Their angles are equal but their sides need not be in the same ratio.
2. $AE = \dfrac{3 \times 6}{4.5} = 4$ cm
3. Yes: $\dfrac{6}{7.5} = \dfrac{4}{5}$
4. $(x+1)^2 = x(x+3)$ gives $x = 1$
5. $QR = 12$ cm, $RP = 15$ cm
6. Yes, SSS: each ratio is $\dfrac{1}{2}$
7. 8 cm
8. 60 m
9. $\dfrac{h}{1.5} = \dfrac{7.5}{4.5}$, so $h = 2.5$ m
10. $\dfrac{3}{5} = \dfrac{4.5}{OD}$, so $OD = 7.5$ cm

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